

# Computer Reviewer

app4r4tu5

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Hello dear reader. Again, welcome back to this reviewer for the 4th Quarterly Examinations for High School. This time, it will involve the more in-depth structures of HTML Multimedia and graphics (although it will be an introductory guide, as most of the questions in the paper will involve those from the textbook). It is more recommended to use the book as most definitions here are *not from the book and are from own thoughts and words*. In the exam, it is most noted that there will be multiple choice and a potential modified true or false, so study the definitions in the book for answering and use this as a supplement to aid understanding what would happen in the exam or whatnot. Also, use the activities done when we were during our asynchronous tasks for better ease as our instructor had quoted using these for the exam's questions. After this short prelude, we shall proceed.

*Created in LaTeX.*

IDE - Visual Studio Code

*"Being the typical makes it more atypical than you think."* - app4r4tu5

I want to give a ♡ for *him*... :,)

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# 1 HTML Multimedia

## 1.1 Illustrations

Added in HTML 5, `<canvas>` is the tag wherein declares area where draw graphics. There are various attributes to go by:

1. `id` - reference name for the script. Their value is alphanumeric.
2. `height` and `width` - creates the canvas size. Their values are numeric.
3. `style` - includes borders<sup>1</sup>

An example on how to use the canvas command, here is a prompt in order to create a gray rectangle:

```
<html>
  <body>
    <canvas id="artwork" width="250" height="200"
      style="border:3px solid black;"> </canvas>
    <script>
      var a=document.getElementById("artwork");
      var atx=c.getContext("2d");
      atx.fillStyle="gray";
      atx.fillRect(12,12,150,75);
    </script> <br>
    This is a square right? It's a quadrilateral!
  <\body>
</html>
```

This will create as seen in the figure on the next page.

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<sup>1</sup>By default, a canvas has no borders.

The figure:

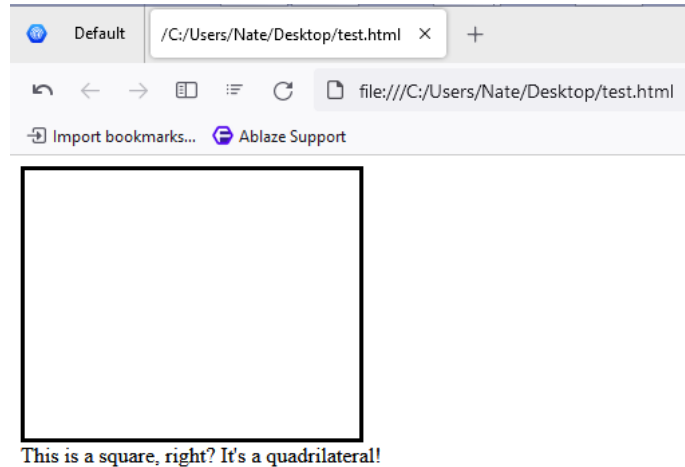


Figure 1: Output of the prompt

We can define various commands that are in the prompt shown a page above:

- `document.getElementById` is the command that calls the canvas title as "artwork."
- `a` is the assigned variable wherein all other commands are dependent on.
- `getContext("2d")` is an object built in HTML5, with several properties and methods for illustrating elements.
- `fillStyle` declares the color, whether solid, gradient, or a pattern.
- `fillRectproperty` uses x, y coordinates, width and height. The x-axis and y-axis assigns the starting point of the image.

#### 1.1.1 Line Paths, Circles, and Text in Canvas

You can also create a line in HTML. There are specific attributes that exist in order to make a line, considered the *line path*.

- `moveTo` is the starting point. `lineTo` is the endpoint of the line.
- `stroke()` is to create the line.

Creating a circle is also simple. It just involves the usage of the command `arc(x,y,r,start,stop)`. `r` is the radius. `start` is the starting point and `stop`

is the endpoint of the circle.

Texts in canvas can be high or no fill, meaning they can have fill in the outline or just have the outline remaining. *Gradients* also apply to each figure, and they can also be linear or radial.

## 1.2 Multimedia

The image format *SVG* or *Scalable Vector Graphics* is a W3C (World Wide Web Consortium) recommended format for defining vector-based graphics in XML formats to be used on web pages. All elements and their attributes in SVG files can be animated. There are advantages to using SVG, such as:

- They can be developed or modified in any *text editor*.
- They can be searched, indexed, and compressed.
- They are scalable to no end, therefore the image would not get pixelated at bigger scales in a web page.
- Printed with high quality at any resolution.
- Zoomable without being pixelated (same as the third bullet).

You can embed audio in an HTML document using a plug-in, like Adobe Flash. However, since browsers have differing structure, each browser may use different plugins to play the audio. The HTML 5 tag `audio` declares a way to drive in an audio file on a web page. Obviously, you must try this out yourself or refer to the book for information as neither the book or this document can play music as a PDF format. The tags for audio are:

1. `<audio>` - declares sound content.
2. `<source>` - calls on multiple media resources for media elements.

There are three formats for audio:

- *MP3* - MPEG<sup>2</sup> Audio Layer 3. Their file extension is most common among sharing audio files, `.mp3`.
- *OGG* - Means "ogging"<sup>3</sup>. Their file extension is `.ogg`.
- *WAV* - an uncompressed audio file developed by IBM and Microsoft. It stands for "Waveform Audio File." Its file extension is `.wav`.

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<sup>2</sup>MPEG stands for "Moving Picture Experts Group"

<sup>3</sup>Ogging means "doing something forcefully, possibly without consideration of the drain on future resources." referenced from the computer game Netrek. Don't ask me, I also don't know.

Chrome 6 and above support all three. Firefox 3.6 and above (according to the book) does not support MP3. Internet Explorer 9 until 11 only supports MP3. Opera 10 and above does not support MP3. Apple Safari 5 and above does not support OGG.

Video can also be seen on web pages, commonly seen in advertisements (albeit just use an adblocker if these are too intrusive). In general, embedding videos not only entertain, but also contributes to the interactivity of the website assuming that the videos are used not as advertisement material. There are video tags denoted for video:

1. `<video>` - defines a video.
2. `<source>` - *Same definition for audio tags.*
3. `<track>` - Defines text tracks in media players.

There are three formats for video:

- *MP4* - MPEG-4. Their file extension is also most common around sharing video files, `.mp4`. MP4 files have an H264 video and AAC audio codec<sup>4</sup>.
- *WebM* - Web Media File. Their file extension is `.webm`. They have a VP8 video and the Vorbis audio codec.
- *OGG* - Same meaning as previously, only that it can also be used for video. They have the Theora video and the Vorbis audio codec.

File support for OGG remains the same among browsers as according to the book. Chrome 6 and above support all three. Firefox 3.6 and above does not support MP4. Internet Explorer 9 until 11 only supports MP4. Opera 10.6 and above does not support MP4. Apple Safari only supports MP4.

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<sup>4</sup>A codec is "a hardware- or software-based process that compresses and decompresses large amounts of data"

## 2 Input Types and Forms in HTML

### 2.1 Input Types

HTML 5 includes features such as input controls and validation that are much more improved than HTML 4 and below. However, some web browsers are yet to support several input controls. The tag `<input>` is used to declare an input control. The list below shows the input types that can be added as in `<input>`, and their support in Chrome and Firefox:

1. **color** - color picker. Only supports Chrome.
2. **date** - date selection. Only supports Chrome.
3. **email** - to input an email address. Supports Firefox and Chrome.
4. **month** - month and year selection. Supports Firefox and Chrome.
5. **number** - numeric values. You can set limits to these with **max**, **min**, **step** as an increment, and the default as **value**. Only supports Chrome.
6. **range** - numeric values from an array of numbers with same attributes as **number**. Only supports Chrome.
7. **search** - an information locator but appears to be a regular text field. Only supports Chrome.
8. **time** - time selection. Only supports Chrome.
9. **url** - web address input. Supports Firefox and Chrome.
10. **week** - week and year controller; time zone doesn't matter. Only supports Chrome.

### 2.2 Forms

In HTML 5, there are three new form elements aside from already introduced.

1. `<input>` and `<datalist>` - identifies a list of pre-defined choices for an output, acts like an "auto-correct." It appears as a drop-down list of already defined inputs as they input data.
2. `<keygen>` - declares a key-pair generator field in a form. It was developed for the purpose of security by authentication of users.

Two keys are created, one private and another public. The private key is saved on the host's computer locally and the public is delivered to the server.

3. `<output>` - specifies the result of a calculation, like computations performed as a script.

### 2.2.1 Attributes of Inputs and Forms

Filling out forms and data needs to be secure and easy. That's why there are newer attributes added to HTML 5 for inputs and forms. Take note that, we will not tackle on how submission of data works, as it is beyond the scope of HTML, as it involves ASP and PHP (research those if you want). *Autocomplete* is applicable both on form and input. When it is turned on, the browser automatically completes values based off previous inputs. They can have autocomplete separated from other tags, on or off.

The **novalidate** attribute expressed that the input *does not need to be verified* in terms of format.

Inputs also have attributes, and they can be placed aside the input types given previously. Those attributes are listed appropriately:

1. **autofocus** - cursor appears at the assigned textbox.
2. **formaction** - the URL that will process the input once submitted.
3. **formenctype\*** - declares the manner of encoding of form data upon submission.
4. **formmethod\*** - specifies the HTTP method for transmission of form-data to the action URL.
5. **formnovalidate** same function as **novalidate**.
6. **formtarget** - denotes a name or a keyword indicating the display location of the response received during form submission.
7. **height** and **width** - exclusive in input type image; sets image size.
8. **multiple** - allows more than one entry.
9. **pattern** - standard value of comparison with respect to the input value; usually applied to text, search, email addresses and passwords.
10. **placeholder** - provides a short clue on the expected value of an input.
11. **required** - declares that a certain input is filled out with information before submission.

\* - can be used along with **type="submit"** and **type="image"**.



*Good luck on the exam dear reader. It would really appreciate if you just read one paragraph, or even one page of this for me to feel better about coding this reviewer for you all. It really makes me feel better knowing that I can help at least by the slightest. Good luck on this examinations, it is the final quarter eitherway. :) - by app4r4tu5.*